

Neurolabs iOS SDK Integration Guide

Technical Integration Specification

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1. Scope

This guide is for partner iOS apps integrating 'NeurolabsSDK' through Swift Package Manager.

2. Changes In v1.1.9

- Default task routing config uses 'taskUUID'.
- Recommended custom-camera parity flow remains strict shelf guidance with overlays disabled.
- 'liveQualityEnabled: true' remains the key switch for pill/rotation/warning/error guidance behavior.
- 'autoCloseAfterCapture: true' with a single required capture remains the recommended parity flow.

3. Requirements

- iOS 17.0+ (SDK package declares '.iOS(.v17)')
- Xcode 16.4+ (release toolchain baseline)
- Swift 5.9+ (Swift 6 toolchains supported)

4. Install (SPM)

Add package dependency (mobile-dist or direct distribution manifest) and link 'NeurolabsSDK'.

5. SDK Initialization + Warmup

```
import NeurolabsSDK

final class CaptureCoordinator {
    let sdk = NeurolabsSDKCore(configuration: .init(
        apiKey: "<API_KEY>",
        apiBaseUrl: URL(string: "https://api.neurolabs.ai")!,
        taskUUID: "<DEFAULT_TASK_UUID>",
        debugLogging: true
    ))

    func prepare() {
        // SDK starts image pipeline preparation during init.
        // Optional: observe loading progress through delegate.
    }
}
```

6. Queue Management

```
Task {  
    await sdk.setAutoSyncEnabled(true)  
    await sdk.setWifiOnlyUploadsEnabled(false)  
    await sdk.setDeletePhotosOnUploadSuccess(true)  
    await sdk.setMaxQueueSize(200)  
    await sdk.setUploadRetryCount(5)  
  
    let status = await sdk.getQueueStatus()  
    print("pending=\(status.pendingCount) failed=\(status.failedCount)")  
  
    await sdk.flushQueue()  
    await sdk.retryFailedUploads()  
}
```

7. Open Custom Camera

```
import UIKit
import NeurolabsSDK

var capture = NLCaptureConfiguration()
capture.confidenceThreshold = 0.25
capture.iouThreshold = 0.45
capture.maxCaptures = 1
capture.guidanceMode = .strict
capture.showDetectionOverlays = false
capture.showCapturedRegions = false

let config = NLCameraConfiguration(
    capture: capture,
    enableValidation: true,
    showAlignmentGuidance: true,
    enableARDetections: false,
    liveQualityEnabled: true,
    liveQualityFPS: 6,
    showCapturePreview: true,
    requiredCaptureCount: 1,
    autoCloseAfterCapture: true,
    sendDetectionsMetadata: true,
    enablePreviewCropping: true
)

let handlers = NLCustomCameraHandlers(
    onSave: { capture in
        // Custom post-processing hook for each accepted capture
        // return .performDefault -> SDK also enqueues upload
        // return .skipDefault -> app takes full ownership
        return .performDefault
    },
    onSaveAll: { captures in
        // Batch hook (multi-capture flow)
        return .performDefault
    }
)

sdk.openCustomCameraUI(
    configuration: config,
    handlers: handlers,
    onClose: {
        print("Camera closed")
    }
)
```

Recommended capture payload notes:

- 'liveQualityEnabled: true' is the key for pill/rotation/warning/error guidance behavior in the custom camera.
- Keep shelf capture mode with strict guidance and overlays disabled for the custom-camera guidance UI path.
- Use 'autoCloseAfterCapture: true' with 'requiredCaptureCount: 1' for the current parity flow.
- 'guidanceMode: .guidance' should only be used as a temporary fallback if a partner explicitly wants looser guidance than the parity path.

8. Post-Processing Hooks

- 'onSave' receives each approved capture.
- 'onSaveAll' receives final batch in multi-capture mode.
- 'onUploadRequested' exists but default queueing is done in 'onSave' / 'onSaveAll' path.

```
let handlers = NLCustomCameraHandlers(  
  onSave: { capture in  
    MyPostProcessor.shared.enqueue(capture)  
    return .skipDefault  
  }  
)
```

9. Delegate/Event Callbacks

```
final class SDKDelegate: NeurolabsSDKDelegate {  
  func neurolabsSDK(_ sdk: NeurolabsSDKCore, didProduceCaptureResult result: NLNativeCaptureResult, for captureId: String, sessionId: String) {}  
  func neurolabsSDK(_ sdk: NeurolabsSDKCore, didEncounterError error: NSError, sessionId: String?, messageId: String?) {}  
  func neurolabsSDK(_ sdk: NeurolabsSDKCore, didChangeQueueStatus status: NLQueueStatus) {}  
  func neurolabsSDK(_ sdk: NeurolabsSDKCore, didUpdateLoadingProgress progress: NLLoadingProgress) {}  
}  
  
sdk.delegate = SDKDelegate()
```

10. Notes

- Per-session routing is supported via native capture config task UUID overrides.
- Ensure 'NSCameraUsageDescription' is present in app 'Info.plist'.